Chapter 1: Foundations of Networking

Lecture notes are based, in part, on (1) slides by Kurose and Ross textbook “Computer Networking: A Top-Down Approach”
(2) Fundamentals of Communications and Networking by M. Solomon and D. Kim

Warm-up questions

Q1: In the Internet model we will primarily use in this course, how many protocol layers are there?

a) 3

b) 4

c) 5

d) 8

Warm-up questions

Q2: In our postal-service analogy, Bob writes the message that he wants to send to Larry. Which Internet layer does Bob most closely represent?

a) Application layer

b) Transport layer

c) Network layer

d) Physical layer

Warm-up Questions

Q3: What is one major advantage of organizing a communication network into layers?

Chapter 1: Goals

- Goal 1: Overview of ***basic concepts***
- Goal 2: Gain familiarity with ***networking terminology***
- Goal 3: Understand the ***Internet structure, protocols, and components***

Chapter 1: roadmap

- Basics of Network Building Blocks and Protocols
- Internet Structure
- Network Core: Packet vs. Circuit Switching
- Network Performance: Delay, Packet loss, Throughput
- Protocol Layers and Service Models

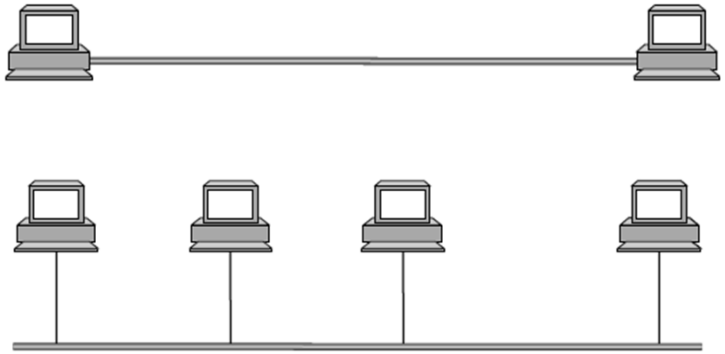
Network Building Blocks

- Nodes:
  - End systems (Computers, servers, printers, …)
  - Switches, routers, …



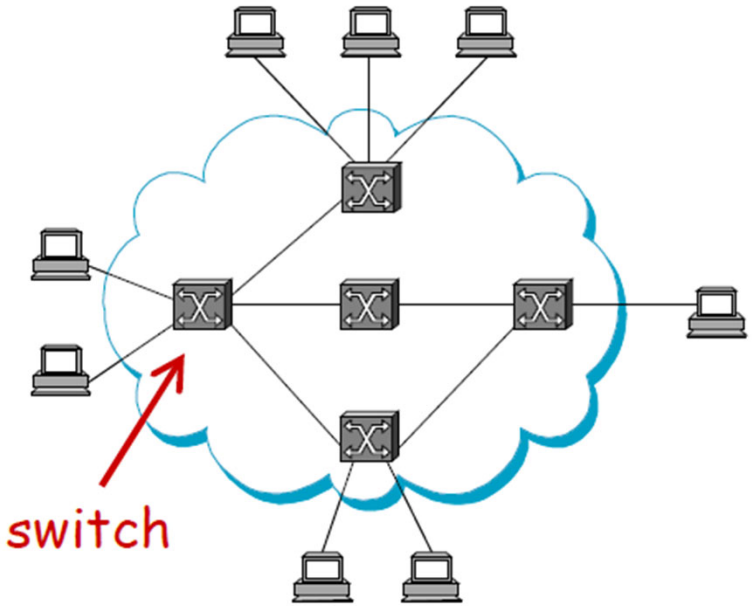


- Links: optical fiber, coax cable,…
  - point-to-point
  - multiple access (*shared media*)



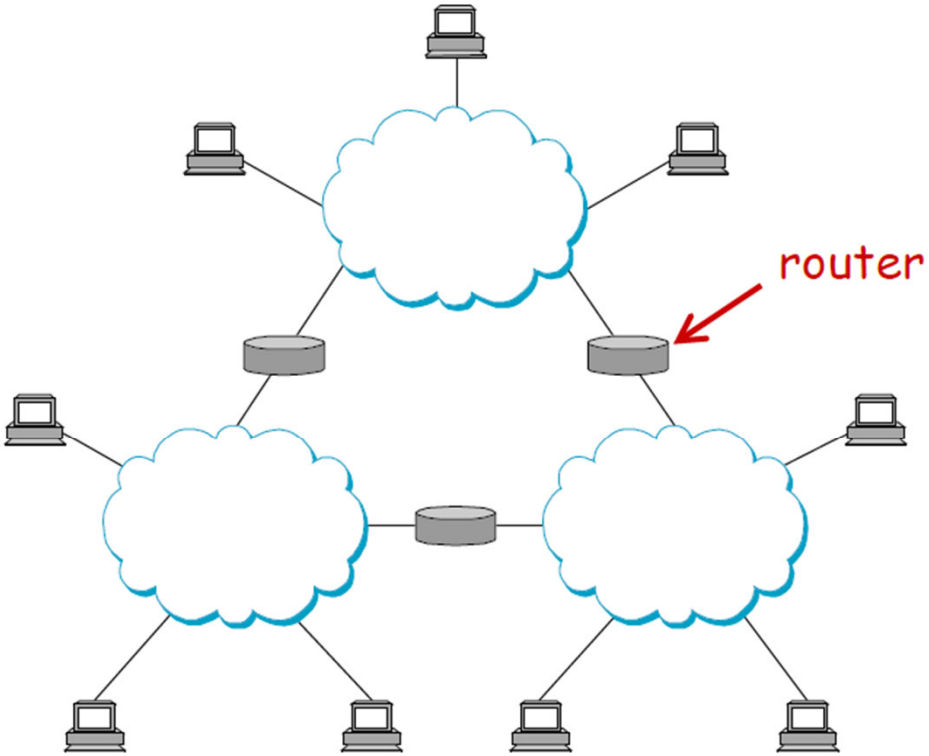
A Network

- A network can be defined as
  - Two or more nodes connected by a link
- End nodes: hosts
- Internal nodes: switches and routers
- ***Primary*** Goal: Provide connectivity
- Other Goals:
  - To meet performance specs
  - Achieve scalability
  - Provide security
  - …



Internetworking

- Connecting networks
- Router: A specialized node that connects between networks
- Requirements:
  - Addressing schemes: IPv4, Ipv6
    - Analogy with ZIP codes
  - Routing protocols
    - Finding end-to-end path from source to destination
  - Forwarding
    - From input interface to output interface



Network Protocols

Human protocols:

- "what's the time?"
- "I have a question"
- introductions

… specific messages sent

… specific actions taken when message received, or other events

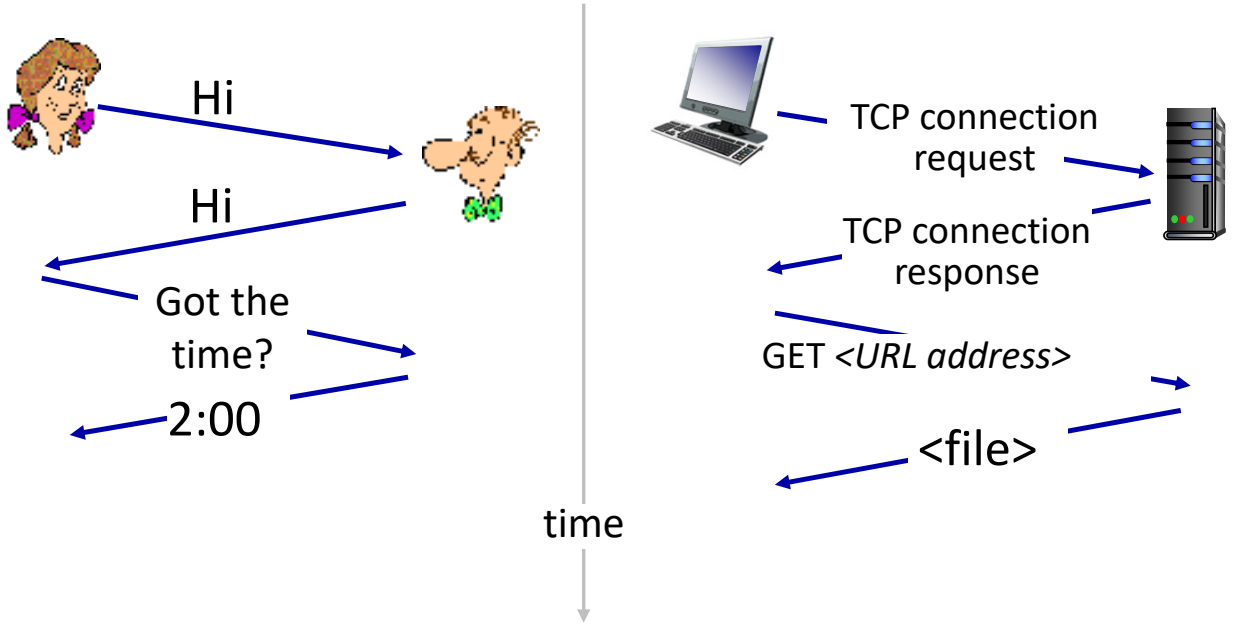
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and actions taken on msg transmission, receipt

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

Chapter 1: roadmap

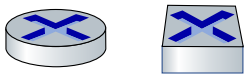
- Basics of Network Building Blocks and Protocols
- Internet Structure
- Network Core: Packet vs. Circuit Switching
- Network Performance: Delay, Packet loss, Throughput
- Protocol Layers and Service Models

The Internet: a "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts = end systems*
- running *network apps* at Internet's "edge"



***Packet** switches*: forward packets (chunks of data)

- *routers, switches*

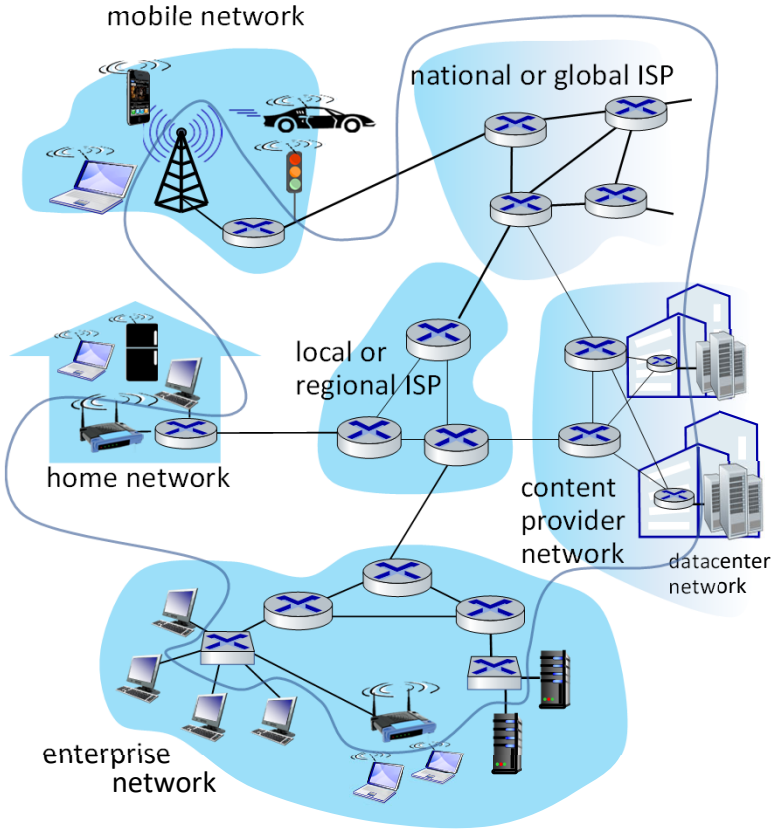


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization

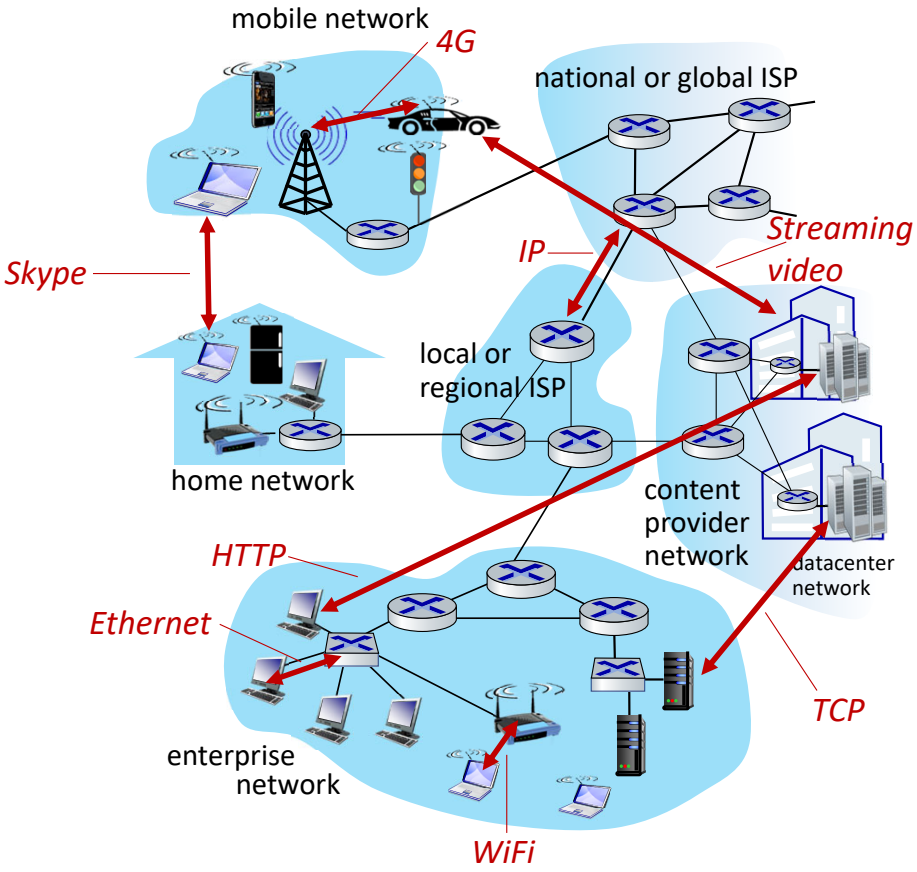


The Internet: a "nuts and bolts" view

- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet

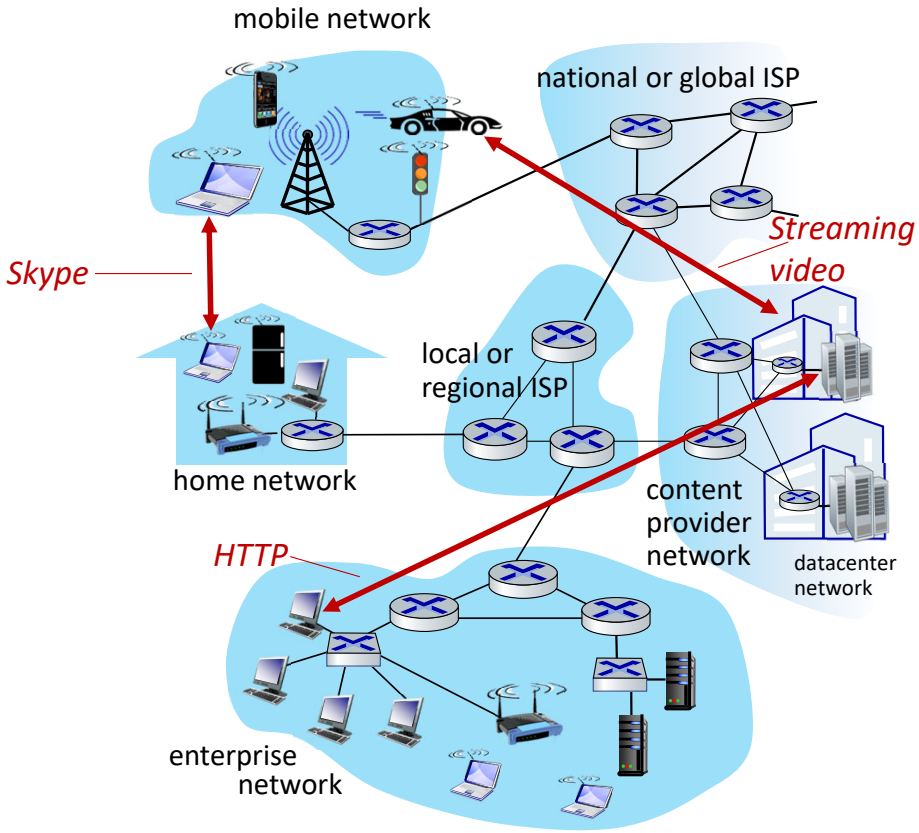
Internet standards

- RFC: Request for Comments
- IETF: Internet Engineering Task Force
- https://datatracker.ietf.org/
- RFC 1 ➔ "Host Software"



The Internet: a "service" view

- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, …
- Provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service



Fun Internet-connected devices ("Service view")



Amazon Echo

Internet refrigerator





IP picture frame



Tweet-a-watt: monitor energy use



Security Camera



Web-enabled toaster + weather forecaster



AR devices

Internet phones





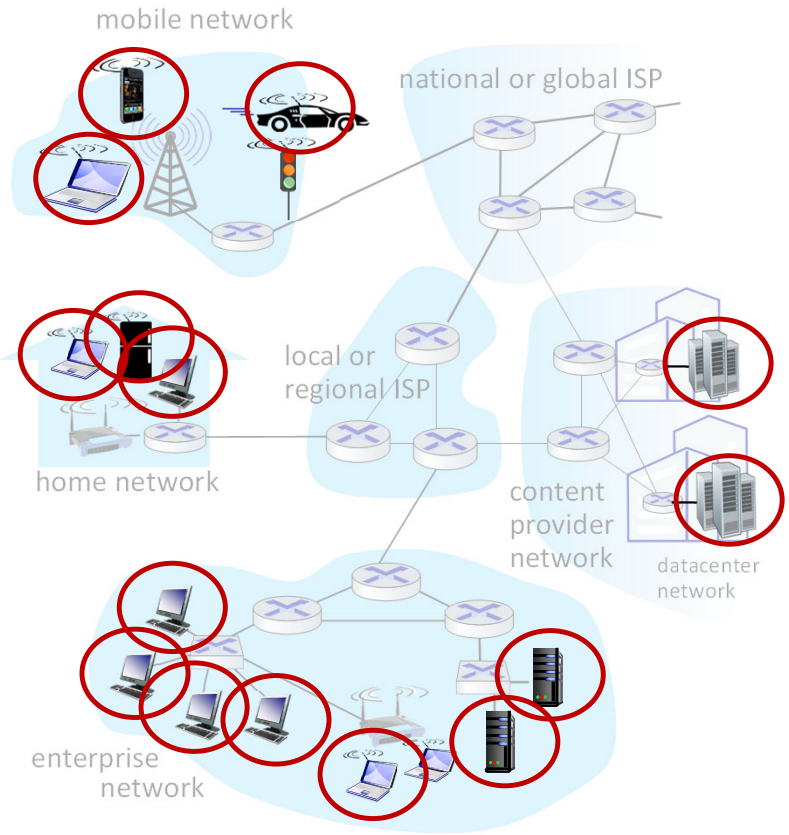
Fitbit

Others?

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



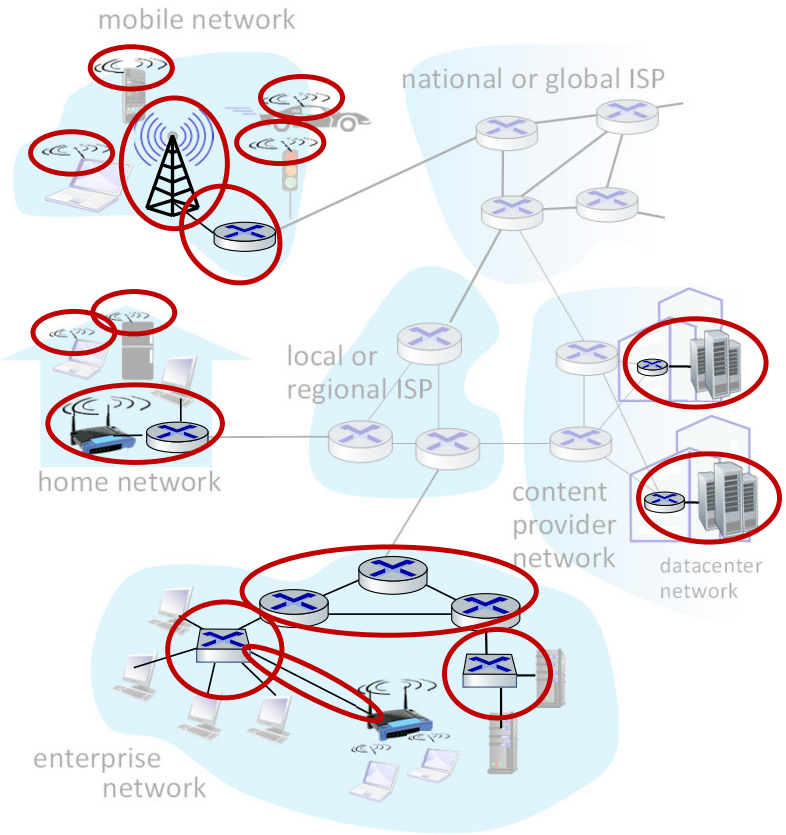
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

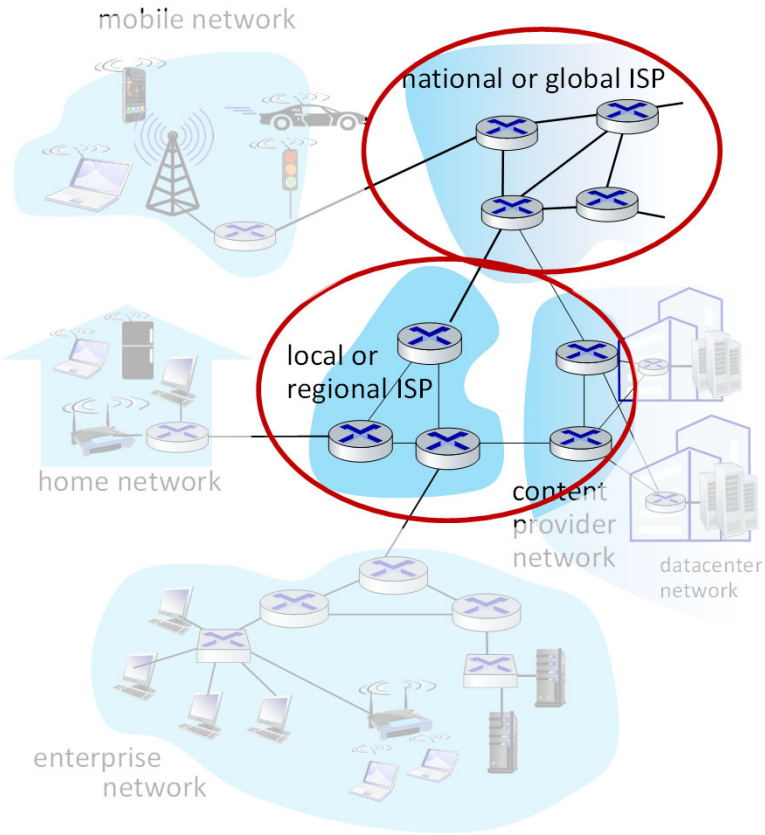
- hosts: clients and servers
- servers often in data centers

Access networks:

- wired, wireless communication links

Network core:

- interconnected routers
- network of networks



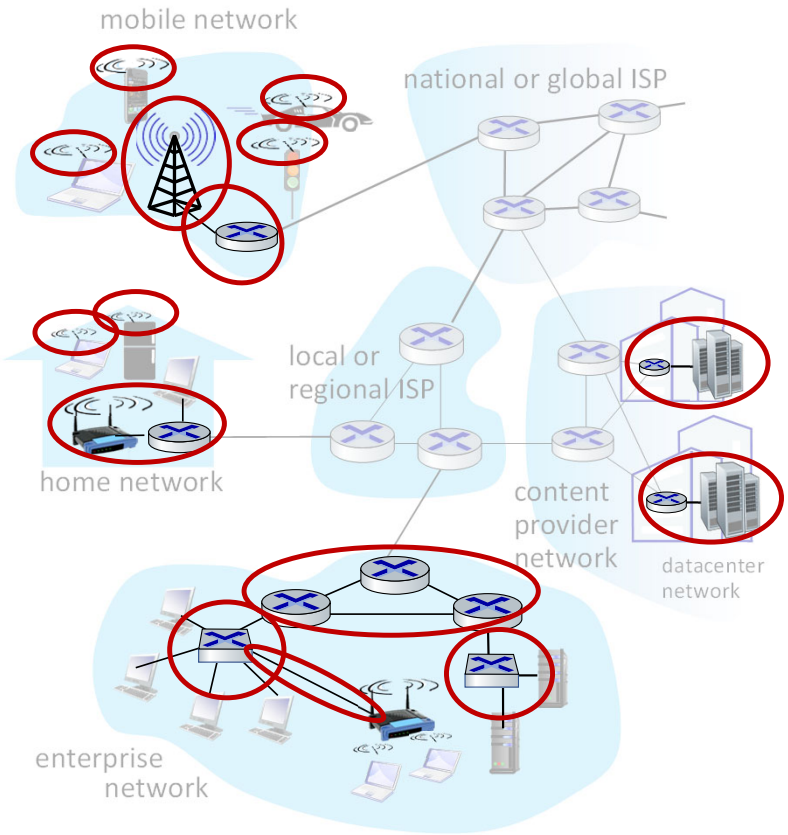
Access networks

Q: How to connect end systems to edge router?

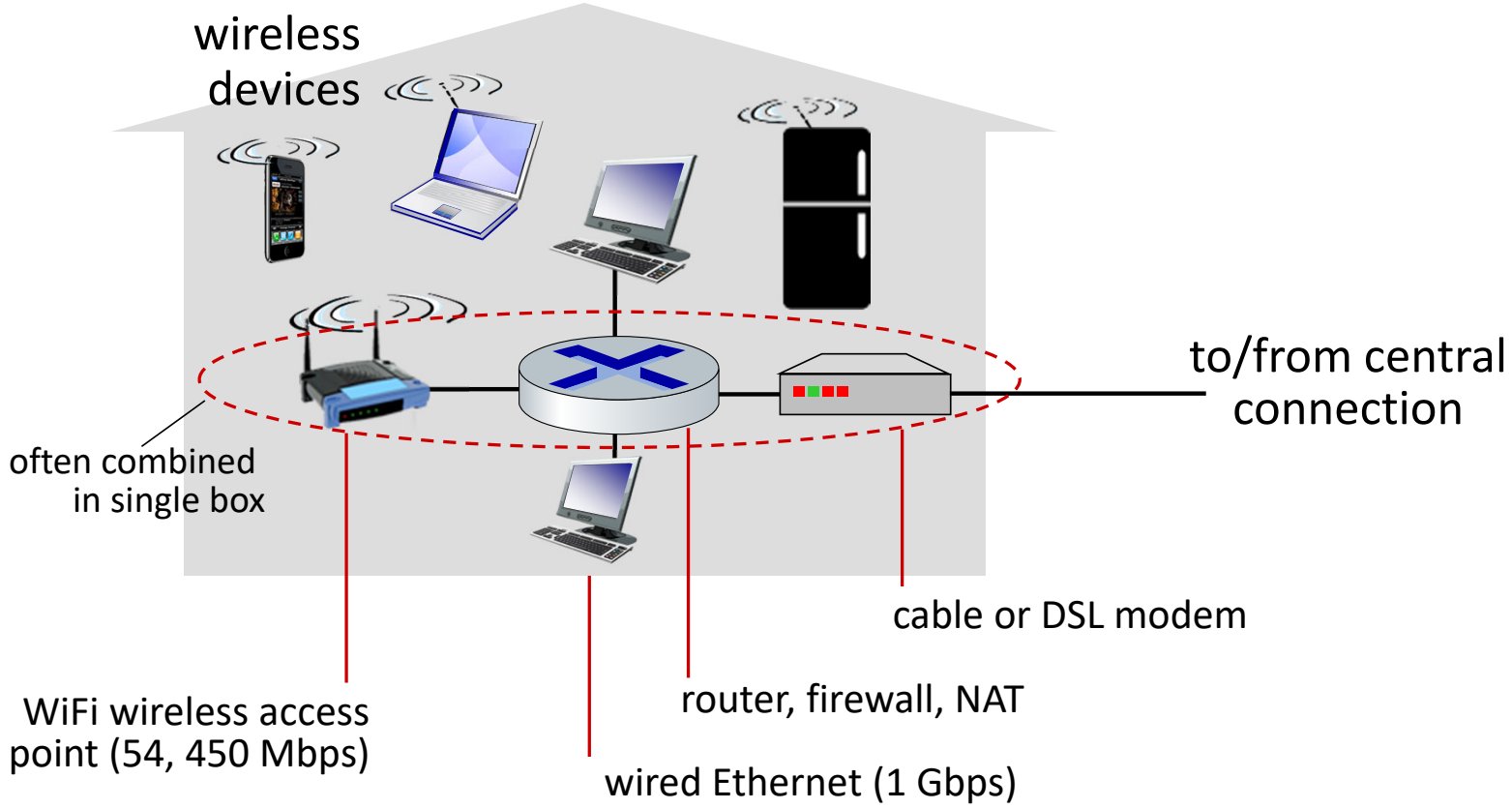
- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

What to look for:

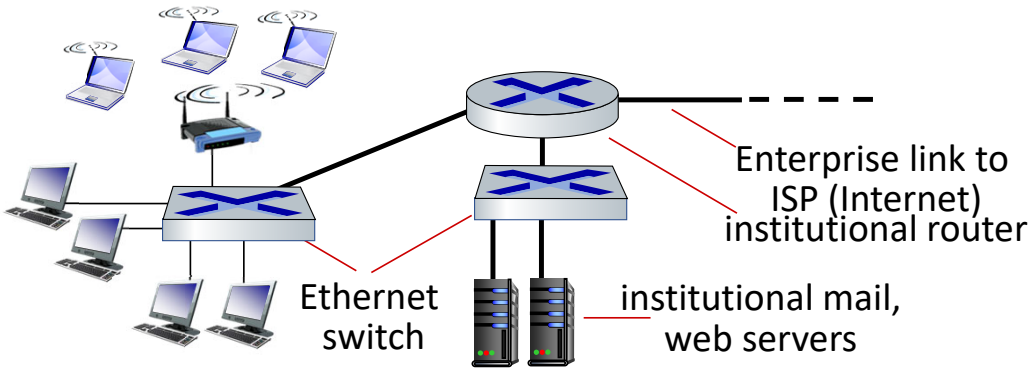
- transmission rate (bits per second) of access network?
- shared or dedicated access among users?



Access networks: home networks



Access networks: enterprise networks



- Companies, universities, etc.
- Mix of wired, wireless link technologies, connecting a mix of switches and routers
  - **Ethernet**: wired access at 100Mbps, 1Gbps, 10Gbps
  - **WiFi**: wireless access points at 11, 54, 450 Mbps
- KU Network Architecture: Link

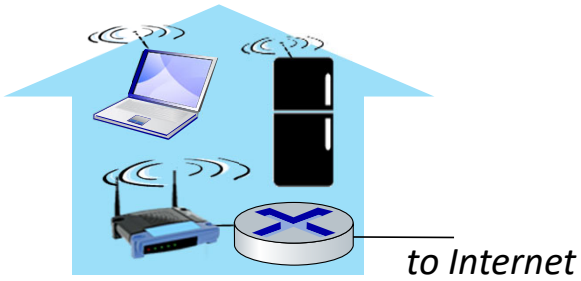
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

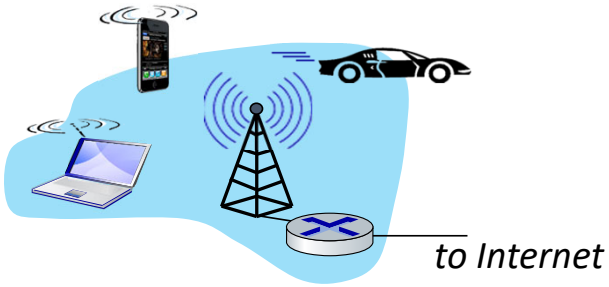
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi)



Wide-area cellular access networks

- provided by mobile, cellular network operator (10’s km)
- 10’s Mbps
- 4G/5G cellular networks (6G coming)



Link Transmission Characteristics

- **Question:** Which of the following technologies has the highest transmission rate in practice?

A. WiFi Links
B. Coaxial cable
C. Satellite Links
D. 4G/5G cellular
E. Fiber optic cable

WiFi Standard

- Let’s do a quick research to learn about different WiFi generations.
- **Question:**
  - Look up the different generations of Wi-Fi (e.g., Wi-Fi 4, 5, 6, etc.).
  - For each generation, identify its corresponding IEEE 802.11 standard

WiFi Standard

Wi-Fi and IEEE 802.11 generations V · T · E

Gen.[1][2]	IEEE standard	Adopt.	Link rate (Mbit/s)	RF (GHz)		
				2.4	5	6
—	802.11	1997	1–2	✓		
	802.11b	1999	1–11	✓		
	802.11a		6–54		✓	
	802.11g	2003		✓		
Wi-Fi 4	802.11n	2009	6.5–600	✓	✓	
Wi-Fi 5	802.11ac	2013	6.5–6,933	[a]	✓	
Wi-Fi 6	802.11ax	2021	0.4–9,608	✓	✓	
Wi-Fi 6E				✓	✓	✓
Wi-Fi 7	802.11be	2024	0.4–23,059	✓	✓	✓
Wi-Fi 8[3][4][5]	802.11bn	TBA		✓	✓	✓

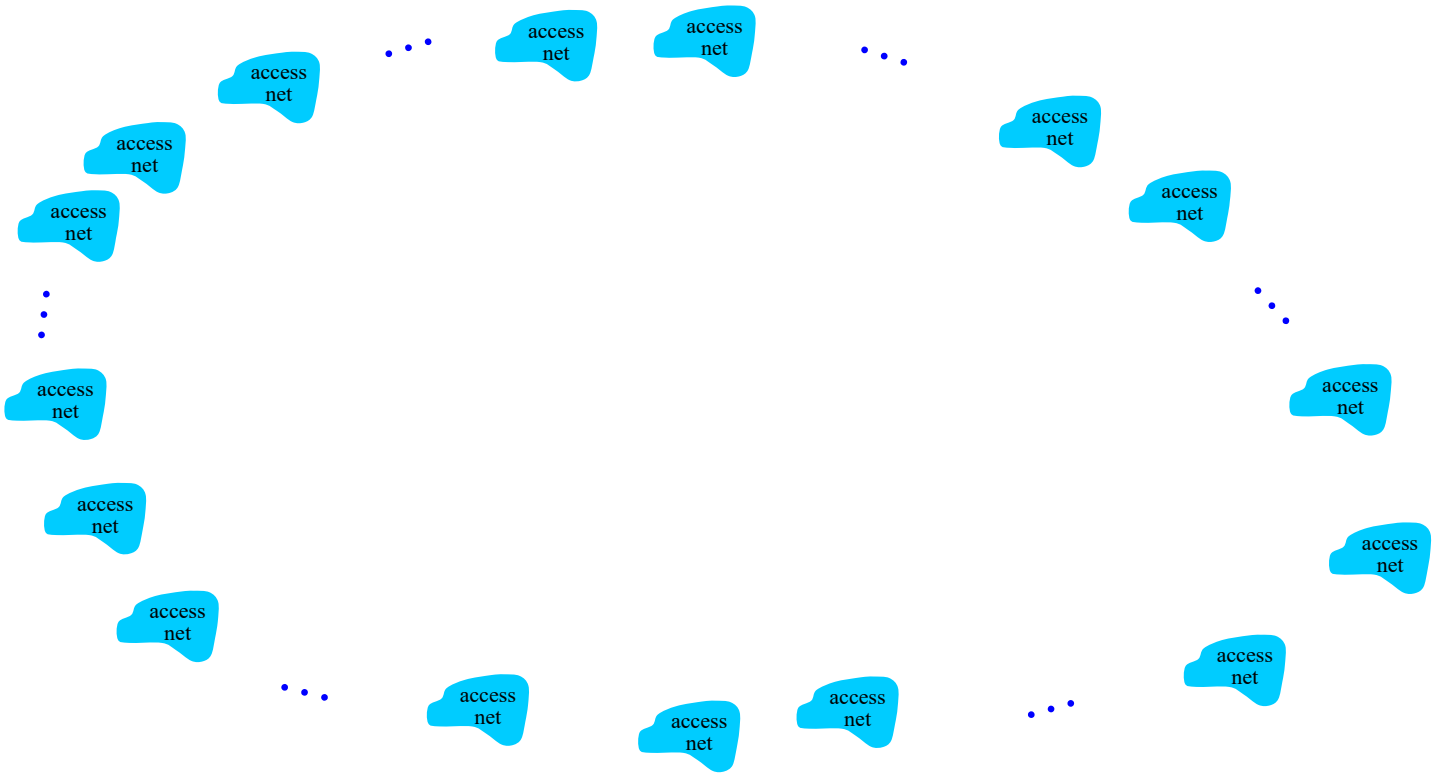
IEEE Standardization Activities: Link

Internet Structure: Network of Networks

- End systems connect to Internet via access ISPs (Internet Service Providers)
  - Residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - So that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - Evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

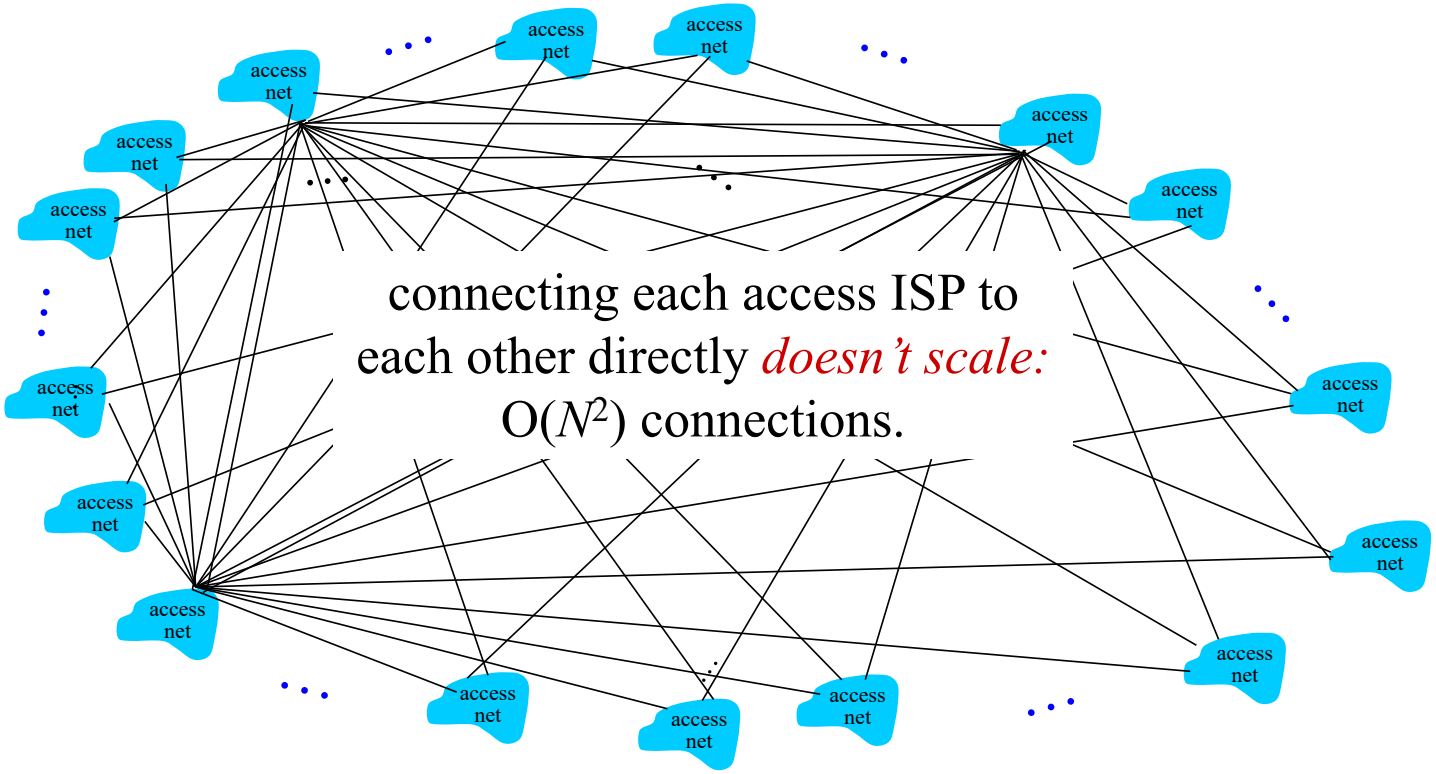
Internet Structure: Network of Networks

Question: given *millions* of access ISPs, how to connect them together?



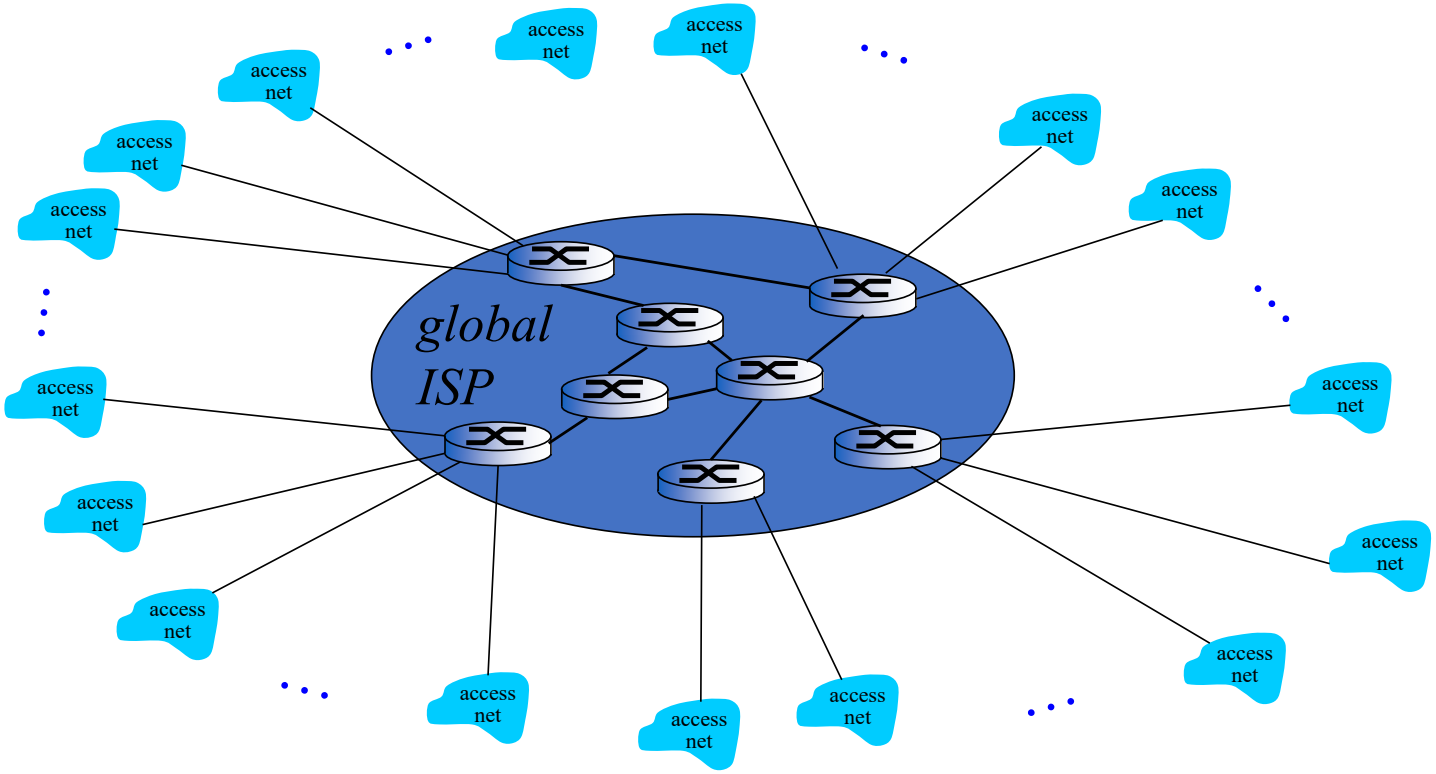
Internet Structure: Network of Networks

Option 1: *connect each access ISP to every other access ISP?*



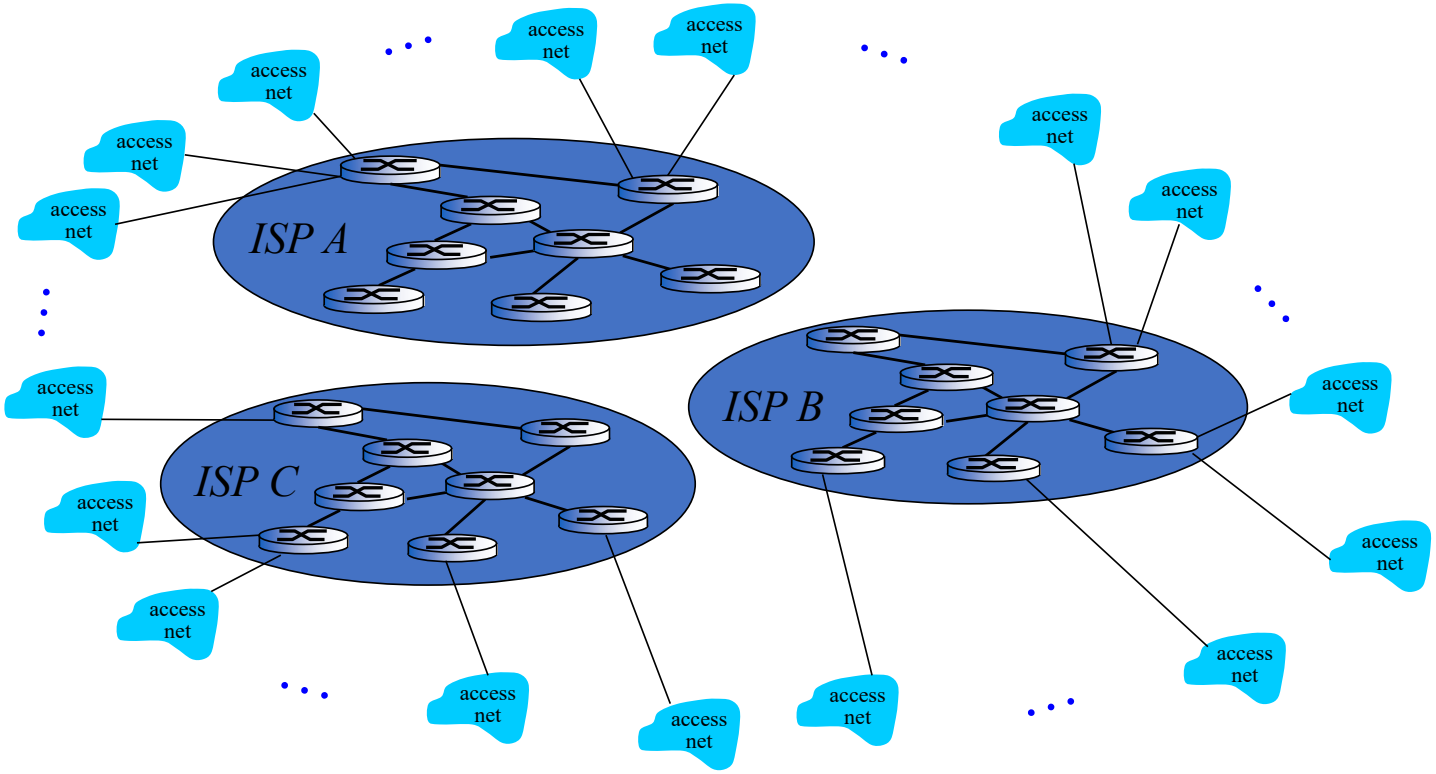
Internet Structure: Network of Networks

Option 2: *connect each access ISP to a global transit ISP?*
Customer and provider ISPs have economic agreement.



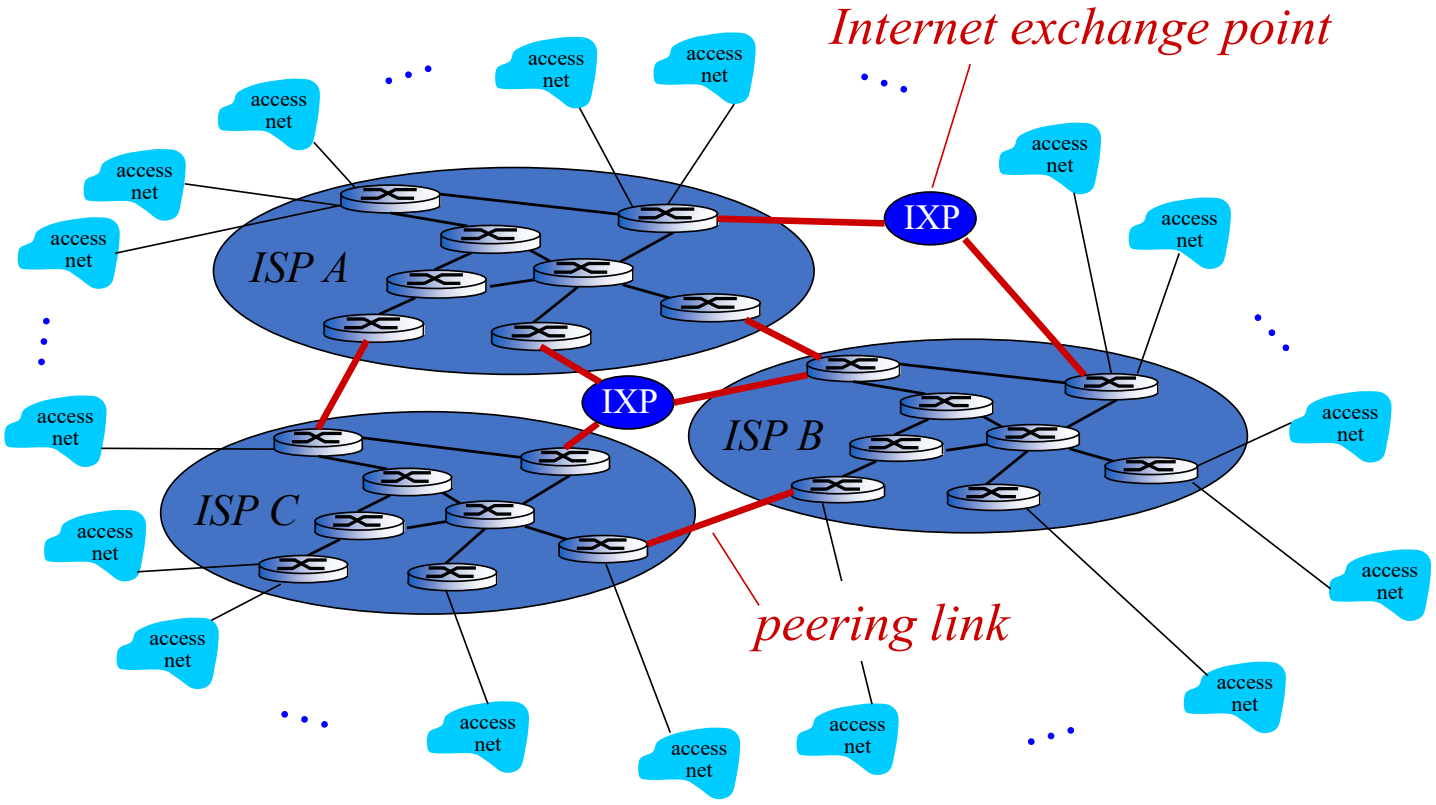
Internet Structure: Network of Networks

But if one global ISP is viable business, there will be competitors ….

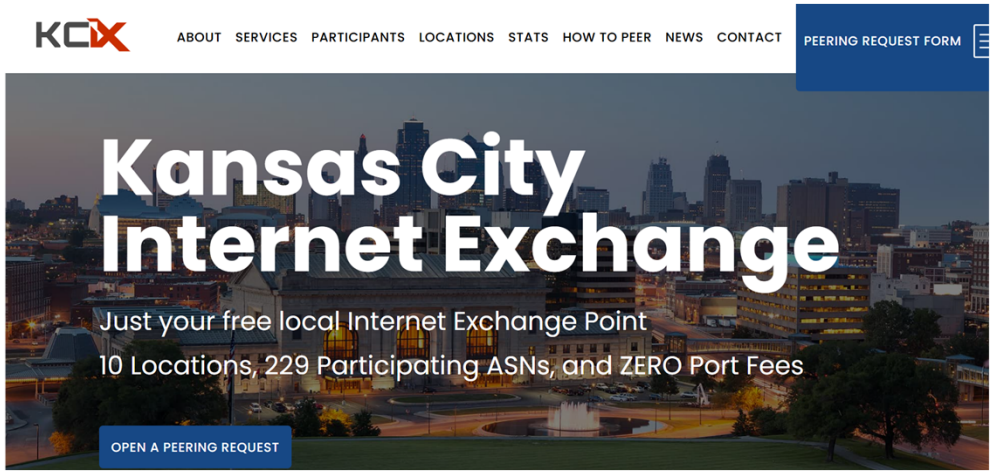


Internet Structure: Network of Networks

But if one global ISP is viable business, there will be competitors …., which must be interconnected.



Internet Structure: Network of Networks



1530 Swift (Nocix): 127 Members

Network	ASN	KCIX IPv4 ▲	KCIX IPv6 ▲	Port	PeeringDB	AS-SET Filter	Peering Policy
Wholesale Internet	32097	206.51.7.1	2001:504:1B:1::1	100Gbps	/NET/3266	AS-WHOLESALE-INTERNET	Open
JCHost	17019	206.51.7.3	2001:504:1B:1::3	1Gbps	/NET/7905		Open
Xtracts, LLC	55241	206.51.7.6	2001:504:1B:1::6	10Gbps	/NET/23762	AS-XTRACTS	Open
SpaceX	14593	206.51.7.7	2001:504:1B:1::7	200Gbps	/NET/18747	AS14593:AS-SX-STARLINK	Open
Unreal Servers	64236	206.51.7.9	2001:504:1B:1::9	1Gbps	/NET/19814		Open
Aaron's Net	25608	206.51.7.11	2001:504:1B:1::11	10Gbps	/NET/24400	AS-AARONSNET	Open
Triplebit	401332	206.51.7.12	2001:504:1B:1::12	10Gbps	/NET/37213	AS-TRIPLEBIT	Open
Google Cache	36040	206.51.7.14	2001:504:1B:1::14	100Gbps	/NET/4319	AS-GOOGLE	Open
PixelDeck Hosting, LLC	40536	206.51.7.16	2001:504:1B:1::16	10Gbps	/NET/39467	AS-PIXELDECK-KC	Open
Cross Computing Services Limited	200639	206.51.7.17	2001:504:1B:1::17	1Gbps	/NET/17628	AS200639	Open
nDeviX	25861	206.51.7.22	2001:504:1B:1::22	1Gbps	/NET/3751	AS25861	Open

Participants:

- Internet Service Providers (ISPs)
- Content Providers and CDNs
- Enterprise Networks (large companies and universities)
- Government and Research Networks (national research and education networks)
- Hosting Providers / Data Centers